CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab Week 7
		Django Models (Part 1)

This lab session covers the basics of Django Models (Part 1). By the end of this session, students should have a very good understanding of how to build a basic model and how to access models in Django.

Pre-lab Preparation:

1. Having bookmodule and usermodule in apps directory, with necessary configurations in settings.py.

Lab Activities:

Task 1: using models, we create a database scheme

ID	Title	Authors	Price	Edition
1	Continuous Delivery	J.Humble and D. Farley	120.00	3
2	Reversing: Secrets of Reverse Engineer	E. Eilam	97.00	2
3	The Hundred-Page Machine Learning Book	Andriy Burkov	100.00	4

Step 1: To have the scheme in the database, we write the following in 'apps/ bookmodule/models.py', Note that: ID is done automatically by Django, so we do not have to explicitly define it in the model.

models.py 2, M

libraryproject > apps > bookmodule > models.py > Book

```python
from django.db import models

class Book(models.Model):
    title = models.CharField(max_length=50)
    author = models.CharField(max_length=50)
    price = models.FloatField(default=0.0)
    edition = models.SmallIntegerField(default=1)
```

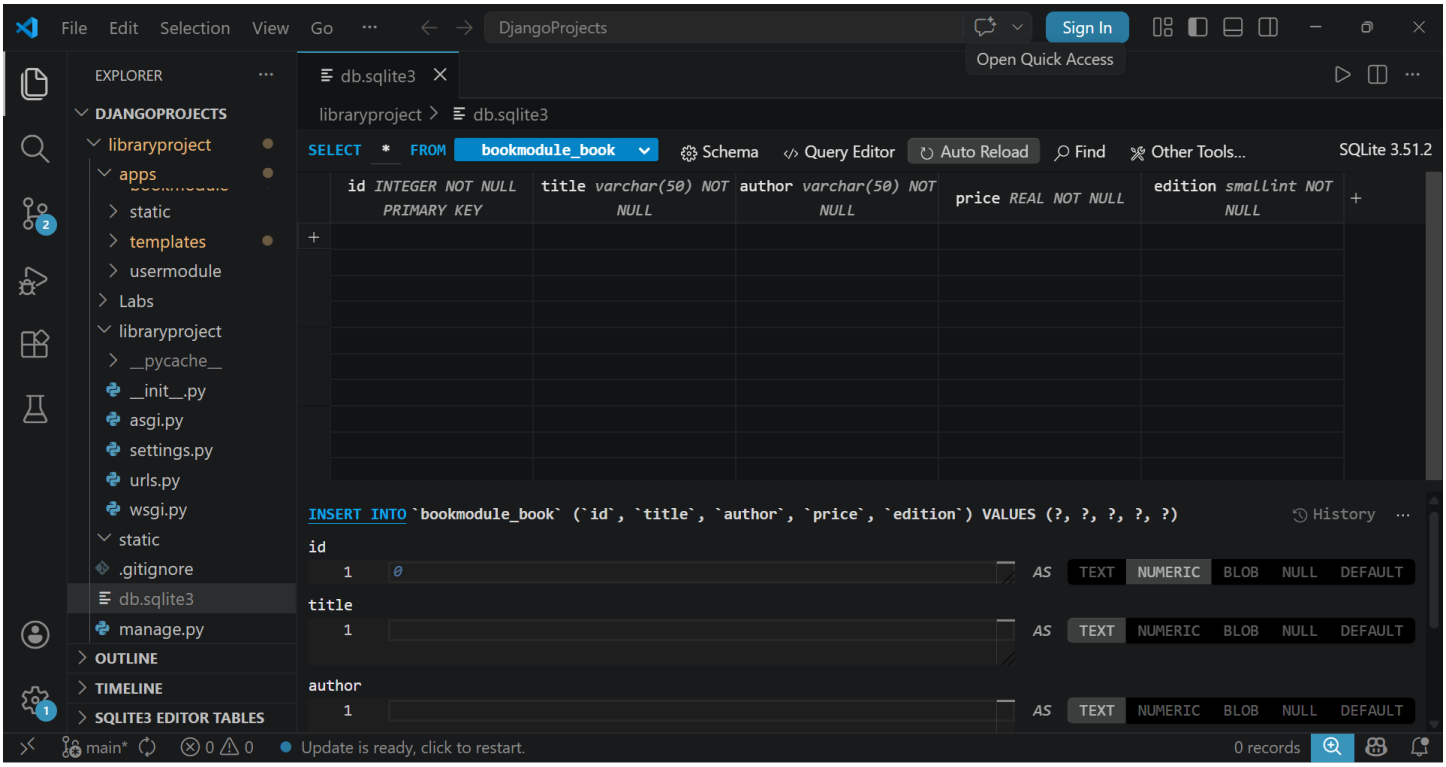
Step 2:&Step 3:

```
(djangoenv) PS C:\Users\re2na\OneDrive\حطب \بتكمل\T_W\practice\DjangoProjects\libraryproject> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, bookmodule, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying bookmodule.0001_initial... OK
  Applying sessions.0001_initial... OK
(djangoenv) PS C:\Users\re2na\OneDrive\حطب \بتكمل\T_W\practice\DjangoProjects\libraryproject>
```

Task 2: use the application 'DB Browser for SQLite' or 'VS code' to view the database and insert fake data into it.

Note: For VS code, you need *SQLite extension*, and for SQLite, you will use *sqlitebrowser* at (https://sqlitebrowser.org/dl/). In the lab, we assumed that you are familiar with at least one of them.

Step 1: Open the database 'db.sqlite3' file in 'VS code' or 'DB Browse for SQLite' apps you can see the table.



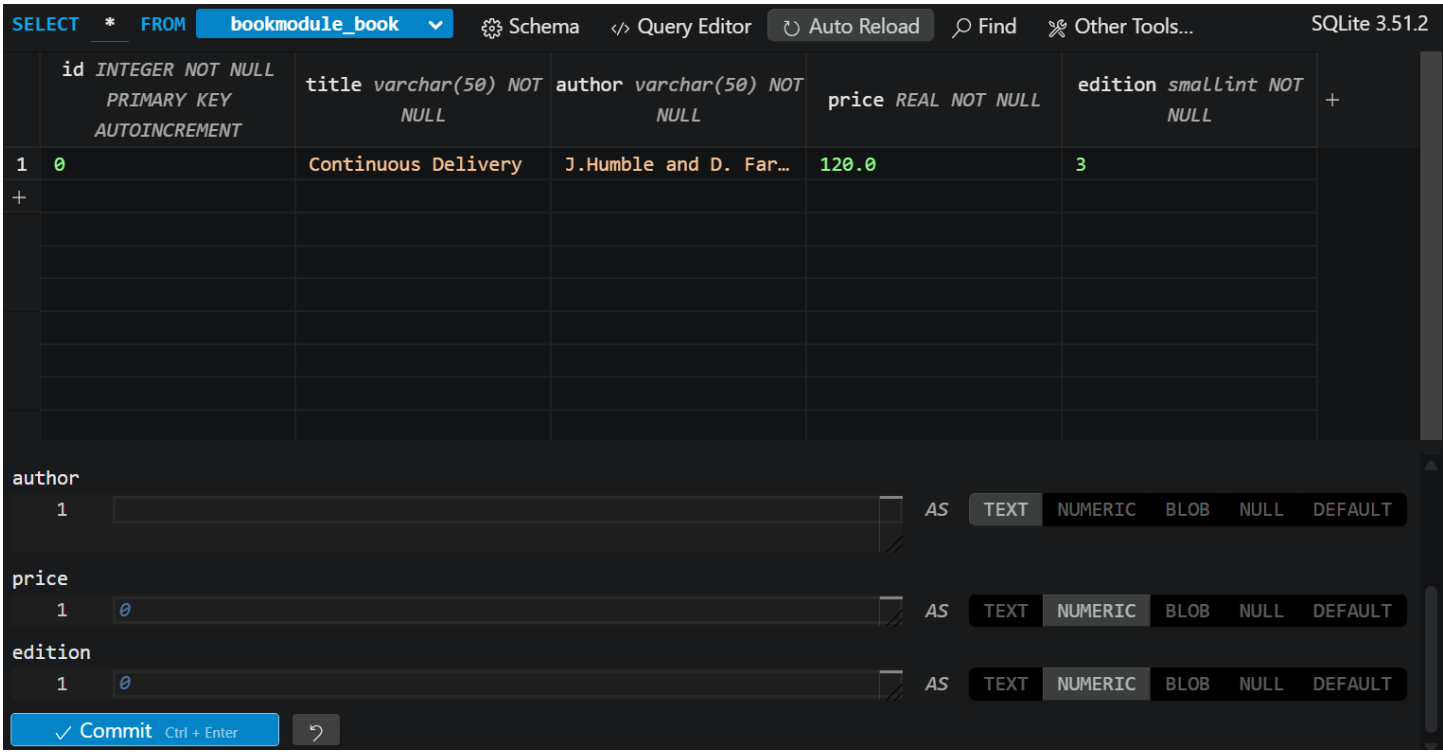
Step 2: For development purposes, you would need data, and you can use random data generator websites, like: www.mockaroo.com

Step 3: Export the data to any file you want; I recommend SQL. Then, import that data into your database. You can use DB SQLite Browser, with the following steps:

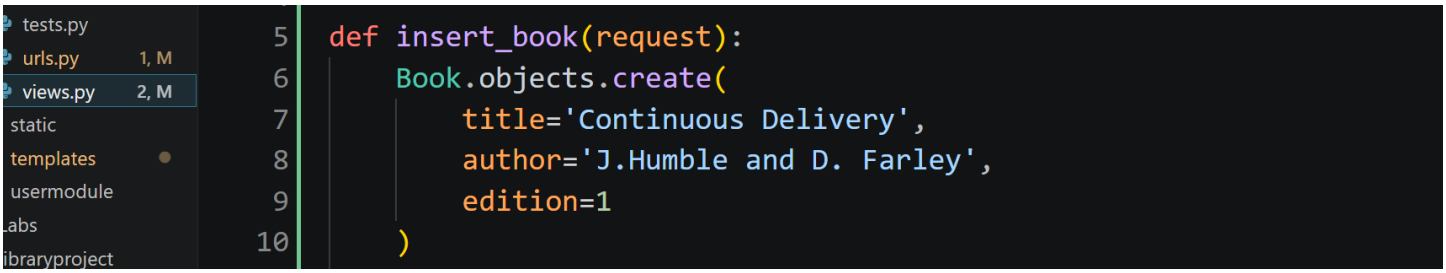
- Open db.sqlite3
- Tab to execute SQL window
- Paste the content of SQL downloaded file into the windows
- Hit execute

Step 4: Also, to insert data from the Django from any view function, you can use the constructor function or create function:

- Use the constructor function



- Use the create function



Either way, we must save the object once created:

```
mybook.save()
```

Task 3: Use simple queries using Django ORM.

Step 1: In 'apps/bookmodule/views.py', import book model:

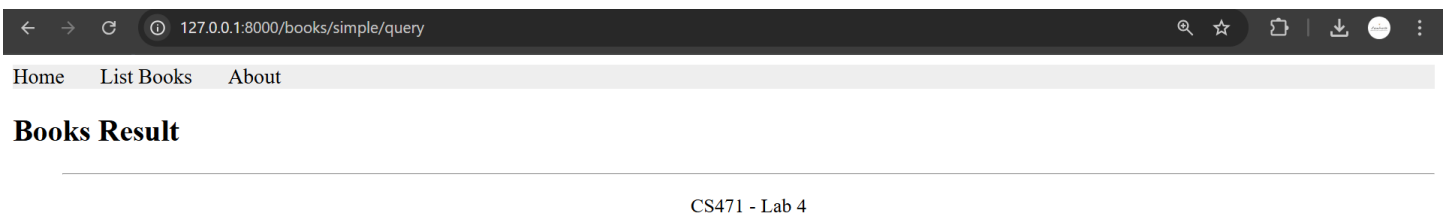
```
from .models import Book
```

Step 2: Create URL '/books/simple/query' with a view function '`simple_query`' in 'apps/bookmodule/views.py':

Step 3: In '`bookmodule/bookList.html`', add Django tag to list books:

```
{% for obj in books %}
    ID: {{obj.id}}, Title: {{obj.title}}
    <hr>
{% endfor %}
```

Step 4: Now from the web browser, navigate to http://127.0.0.1:8000/books/simple/query



Task 4: Customize queries using common lookup conditions and keywords.

Step 1: Create URL '/books/complex/query' with a view function '`lookup_query`' in 'apps/bookmodule/views.py':

```
def complex_query(request):
    mybooks=books=Book.objects.filter(author__isnull =
False).filter(title__icontains='and').filter(edition__gte = 2).exclude(price__lte = 100)[:10]
    if len(mybooks)>=1:
        return render(request, 'bookmodule/bookList.html', {'books':mybooks})
    else:
        return render(request, 'bookmodule/index.html')
```

Step 2: Now from the web browser, navigate to http://127.0.0.1:8000/books/complex/query

Home List Books About
ID: 2, Title: Networks and Security

Books Result

ID: 2
Title: Networks and Security
Author: Ali Ahmad